Chapter 1

Management of dental caries

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DENTAL CARIES THE DISEASE

Dental caries is a disease that is common to all dentate individuals. At the hydroxyapatite crystal level it could be considered a ubiquitous phenomenon. For dental caries to occur a bacterial biofilm has to accumulate on a tooth surface. The bacteria within the biofilm metabolize dietary sugar substrates producing acids which, over time, lead to demineralization of the tooth tissue. Thus the requirements for the carious process may be depicted by the Venn diagram seen in Figure 1.1. However, this diagram is overly simplistic and implies that the disease process and its progression are inevitable. Clinically, this is not the case. Some tooth surfaces that are frequently covered in plaque do not develop clinically detectable caries whereas other tooth surfaces covered with plaque in the same mouth do. Many other factors, such as dietary habits, fluoride and saliva impact upon the disease process which is complex and dynamic in nature. From the earliest stage, continued demineralization is not inevitable and lesion arrest is possible by simply disrupting the plaque biofilm on the surface of the tooth at regular intervals. Very early lesions which are not detected clinically may therefore not progress to clinically detectable white spot lesions and the carious process is better represented by Figure 1.2.

It is often said that if an early white spot lesion develops, remineralization can take place. While complete remineralization and resolution of the lesion is unlikely, the clinically apparent remineralization of a white spot lesion may also be due to its surface abrasion following improved oral hygiene procedures. Remineralization or regrowth of partially demineralized apatite crystals in the surface zone of an enamel lesion (Figure 1.3) has been reported. However, the relatively well-mineralized surface zone acts also as a diffusion barrier to ions, making it less likely that, in the underlying body of the lesion, supersaturation with respect to apatite will occur with subsequent mineral deposition. Prevention of the disease is therefore our primary aim. However, for many patients, this primary prevention fails and lesions develop. Caries risk assessment and early caries detection are important so that further prevention can be targeted to those patients and lesions that are in need. In this situation a method of monitoring the lesion is also important to determine the outcome of our preventive approach; that is, has the lesion arrested or progressed.

Primary prevention can fail for a number of reasons. It may be due to the fact that the patient has not visited

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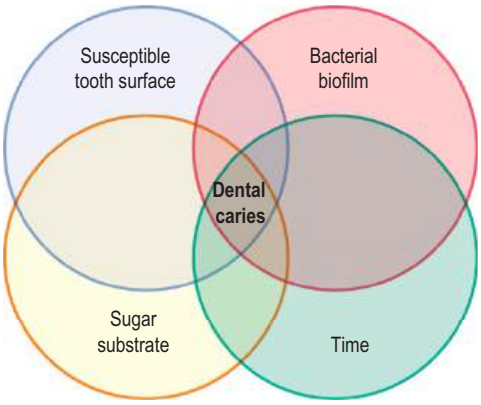


Figure 1.1 Venn diagram depicting the requirements for caries to occur.

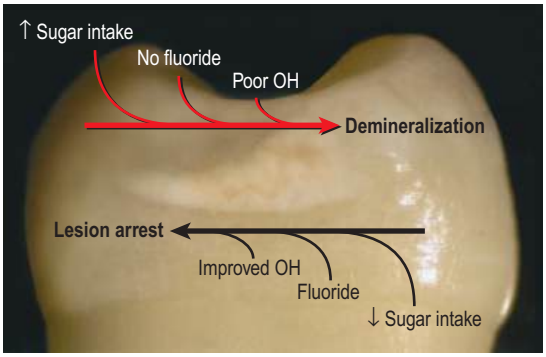


Figure 1.2 Diagram depicting the dynamic nature of the carious process and how it can be influenced by some external factors.

a dentist to receive such advice or, worse still, they may have attended a dentist or dental care professional and not been given preventive advice and treatment. However, for some patients, regardless of their attendance pattern, the preventive advice is ignored or they are unable to follow it through no fault of their own. For example, an elderly patient may know that oral hygiene procedures are important in caries prevention, but they may not have the manual dexterity, due to a physical disability, to carry them out efficiently.

When primary prevention fails, demineralization within the carious lesion can progress to a stage that it becomes heavily infected with bacteria and no longer manageable with preventive procedures alone. Indeed, the surface of the lesion can break down and a microcavity or frank cavity can result which can no longer be kept clean of plaque. In these situations caries removal and restoration with an appropriate dental material is required. The patient has now entered the restorative cycle.

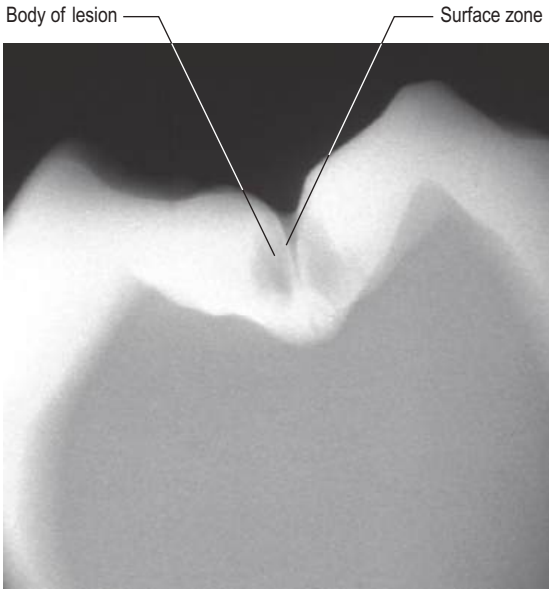


Figure 1.3 High definition macroradiograph of a section of a tooth showing two enamel lesions on the bilateral walls of a fissure. The relatively well-mineralized surface zone and body of the lesion can clearly be seen.

THE RESTORATIVE CYCLE

A significant proportion of a dentist's work time is spent replacing restorations and the most common reason given to justify this clinical decision is the presence of secondary caries. This is caries that develops under or adjacent to a restoration placed to repair a carious lesion. The term secondary caries is, however, misleading as it implies that the restoration is somehow the cause. In certain situations this is true when the restorative procedures have been carried out incorrectly. For example, ledges create plaque stagnation areas, poor contact points allow food packing and poor adaptation with an inadequate bond of materials to tooth tissue leads to microleakage (Figure 1.4). Most 'secondary caries' is actually new caries that has just formed adjacent to the restoration and is better termed as such: caries adjacent to a restoration (CAR). The appearance of caries, following restoration of a tooth, illustrates the continued high caries risk of the patient and also shows that restoration alone does not change this. In order to prevent recurrence it is fundamental that the caries risk be managed. If successful, this might avoid the need for more advanced restorative work, and ensure its predictability when carried out.

Unlike directly placed restorations, which are packed and adapted to cavity walls and margins, indirect restorations are made on models from impressions taken of

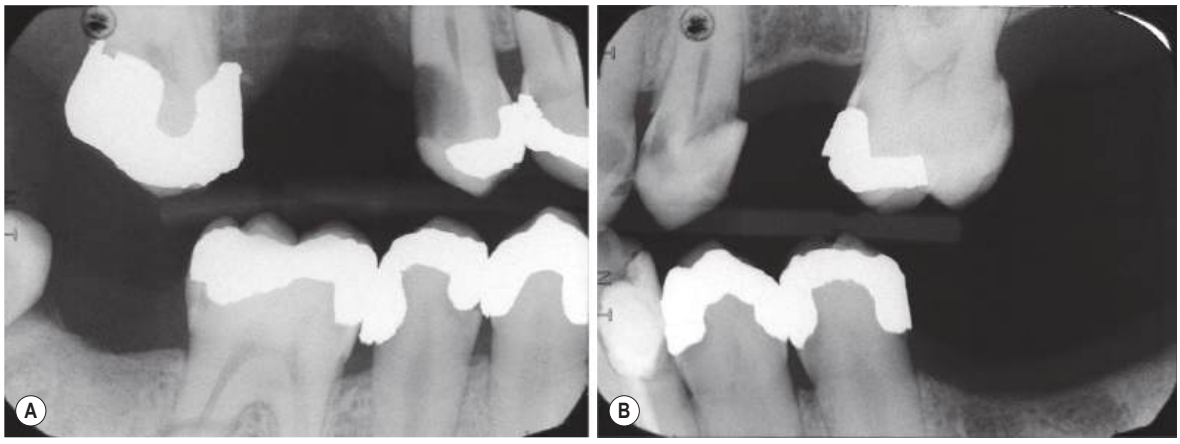


Figure 1.4 Bitewing radiographs of a high caries risk patient. The amalgam restorations placed in the lower right second premolar tooth distally, the upper right second molar mesially and the composite restoration upper left first premolar distally have ledges and are poorly contoured, encouraging food packing and caries adjacent to the restorations.

tooth preparations. As such, discrepancies in the marginal fit, seating and hence the width of cement lute exposed to the oral cavity can occur. Marginal discrepancies in the order of 70 μm have been reported in well-fitting restorations. When fitting an indirect restoration it is important to assess its fit to ensure the marginal discrepancies are kept to a minimum. A dental probe is useful for checking this and for ledges. Poor plaque control in relation to ill-fitting and contoured restorations not only increases the patient's risk to new caries, but also to periodontal disease if the margins are close to the gingival tissues.

CARIES DETECTION AND DIAGNOSIS

It is important when examining a patient for primary caries or caries adjacent to restorations that the teeth are examined clean. The carious process initially takes place in the biofilm on the surface of the tooth and the product of that process is the initial lesion in the tooth. To see the lesion and make a diagnosis the biofilm needs to be removed (Figure 1.5). It is also essential that the teeth are examined both wet and dry. The importance of drying is illustrated in Figure 1.6. When light illuminates a sound tooth, the light can either be transmitted, or it can undergo refraction or reflection. Refraction is the ability of a tooth to bend (scatter) light and will vary according to the refractive index of the material the light passes through. The porosities created in enamel during the carious process are normally filled with water (refractive index = 1.33) which has a refractive index close to enamel (1.66). In this situation little light scattering occurs. If the lesion is dried and the water is replaced with air which has a lower refractive index (1.0), the larger difference in refractive indices between enamel and air results in greater light scattering, enabling easier recognition of the white spot lesion. The occlusal lesion in Figure 1.7 clearly illustrates this and it stands to reason that a lesion that needs to be dried to enable its diagnosis is less severe than one that is seen even on a wet surface. The examination of clean teeth under both wet and dry conditions forms the basis for a clinical visual classification system known as the International Caries Detection and Assessment System (ICDAS II). This system characterizes lesions of increasing severity by correlating the visual appearance of the lesions with their histological depth (Table 1.1). The ICDAS II criteria can also be applied to caries adjacent to restorations. For further information on the ICDAS, visit the website http://www.icdas.org/index.htm.

A number of caries detection devices have been invented to aid detection and monitoring of early carious lesions; however, their use has mainly focussed on primary caries.



Figure 1.5 Disclosed biofilm on the buccal surface of the lower left second premolar tooth has partially been removed exposing the white spot lesion beneath. The biofilm on the mesial surface of the first molar tooth completely obscures the detection of the white spot lesion beneath.

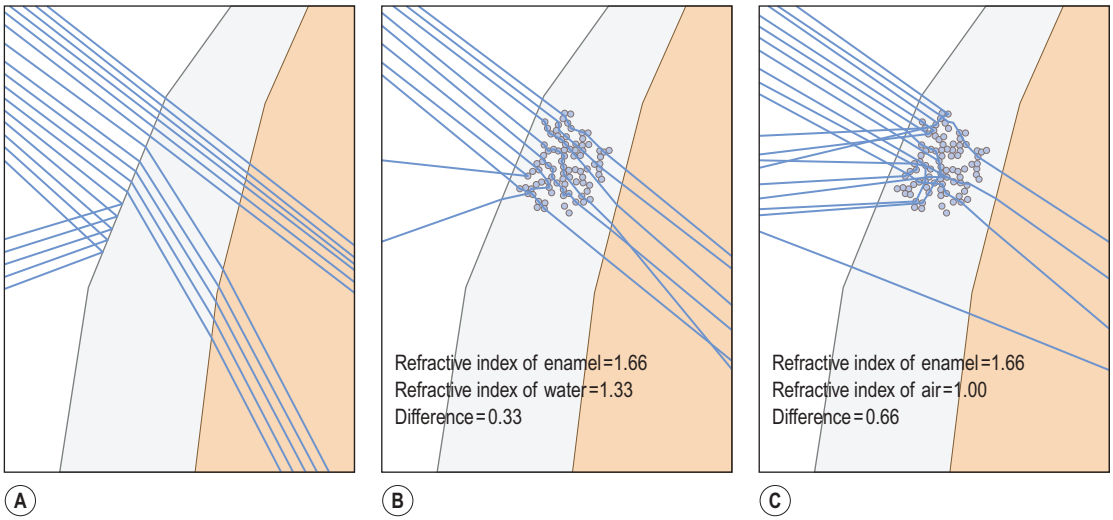


Figure 1.6 When light illuminates sound enamel (A) it undergoes transmission, refraction and reflection, porosities created during the carious process are normally filled with water (B) with a refractive index close to that of enamel. As a consequence, there is little light scattering. When the lesion is dried and air fills the porosities (C) the difference in refractive index with enamel is greater and greater light scattering occurs, making the white spot lesion easier to detect.

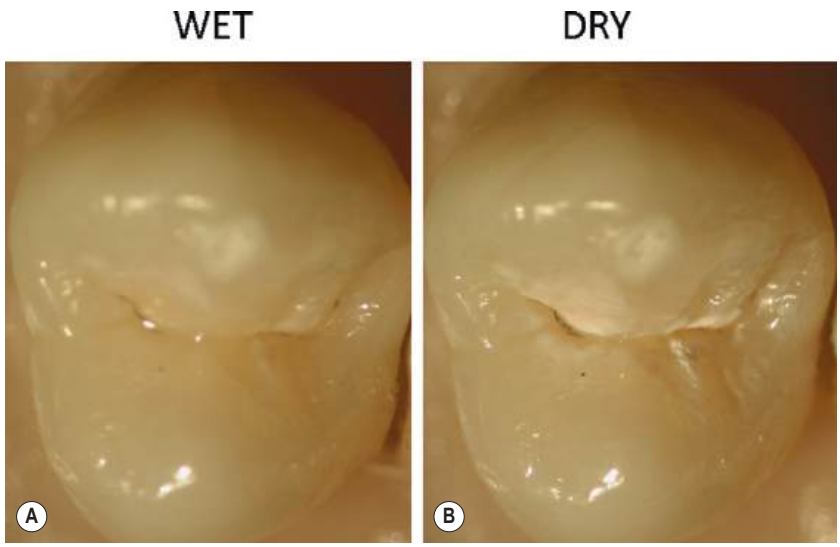


Figure 1.7 White spot lesion at entrance to the fissure (A) is seen more clearly when dry (B).

When a restoration is placed their use is often complicated; for example, the laser fluorescence devices, the DIAGNOdent and the newer DIAGNOdent pen (Figure 1.8), which detect fluorescence from bacterial porphyrins, cannot differentiate between staining around a restoration margin and caries as they both fluoresce. Similarly, false-positive readings could occur if electrical conductance methods were used around metallic restorations. The clinical visual examination, supplemented with an intraoral radiograph, remains the examination of choice for the evaluation of restoration margins and adjacent caries. A dental probe can be used to remove plaque, assess the fit of a restoration and assess for any loss in tooth surface integrity (cavitation), but it must not be used with pressure to detect stickiness as this can lead to errors in diagnosis and more importantly damage to early lesions.

Table 1.1 International Caries Detection and Assessment System (ICDAS II)[a]

ICDAS CRITERIA	COLLAPSED CRITERIA FOR CLINICAL USE	CORRESPONDING HISTOLOGY
0 No or slight change in enamel translucency after prolonged air drying (>5 s).	**0** No or slight change in enamel translucency after prolonged air drying (>5 s).	**0** No enamel demineralization or a narrow surface zone of opacity (edge phenomenon).
1 Opacity or discoloration hardly visible on a wet surface, but distinctly visible after air drying.	**1** Opacity or discoloration hardly visible on a wet surface, but distinctly visible after air drying.	**1** Enamel demineralization limited to the outer 50% of the enamel layer.
2 Opacity or discoloration distinctly visible without air drying.	**2** Opacity or discoloration distinctly visible without air drying.	**2** Demineralization involving between 50% of the enamel and the outer third of dentine.
3 Localized enamel breakdown in opaque or discoloured enamel. **4** Greyish discoloration from the underlying dentine.	**3** Localized enamel breakdown in opaque or discoloured enamel and/or greyish discoloration from the underlying dentine.	**3** Demineralization involving the middle third of dentine.
5 Cavity in opaque or discoloured enamel exposing the dentine – involving less than half of the tooth surface **6** Cavity in opaque or discoloured enamel exposing the dentine – involving > half tooth surface.	**4** Cavity in opaque or discoloured enamel exposing the dentine.	**4** Demineralization involving the inner third of the dentine.

[a]The original codes and description of lesions are seen in the left column, a collapsed version more appropriate for clinical practice in the middle and the corresponding lesion depth as would be seen histologically on the right.

The identification or detection of a carious lesion as outlined here is only one part of the diagnostic process. Lesion characteristics, including severity (depth and mineral loss) and activity, as well as caries risk factors (see later), are all taken into consideration. It is only when all this information is processed that a true diagnosis, prognosis and treatment plan can be made.

CARIES RISK ASSESSMENT

The patient seen in Figures 1.4 and 1.9 is a young patient who has received multiple restorations, a number of endodontic treatments and several extractions as a result of caries. There is new primary caries and caries adjacent to existing restorations which has led to loss of restorations and tooth fracture. This patient has a high risk of developing caries unless a treatment plan dominated by a preventive approach is adopted. Simply re-restoring the teeth is not addressing the caries problem.

What is caries risk?

Caries risk is a prediction as to whether a patient is likely to develop new caries in the future. This is a complex process, which a dentist does on a day-to-day basis, by either consciously or subconsciously assessing the impact of factors which affect caries. It is important to note that caries risk can change throughout a patient's life and a dentist can have a positive impact on reducing this risk. Events in a patient's life can also have a negative effect on caries risk; for example, an elderly patient may be placed on medication which causes a dry mouth. In addition, this patient may require a removable partial prosthesis which will complicate oral hygiene procedures.

For the patient in Figures 1.4 and 1.9, extensive and advanced restorative procedures are likely to be necessary. But before they can be considered his caries risk has to be modified. Even though it is obvious that he is currently at a high caries risk, an assessment should be made to establish the main risk factors and a treatment plan formulated to address them. Once this preventive approach has been instituted, stabilization of active caries is required and



Figure 1.8 The DIAGNOdent pen laser fluorescence device (A) with display on the back of the handle (B) and the approximal sapphire tip (C).

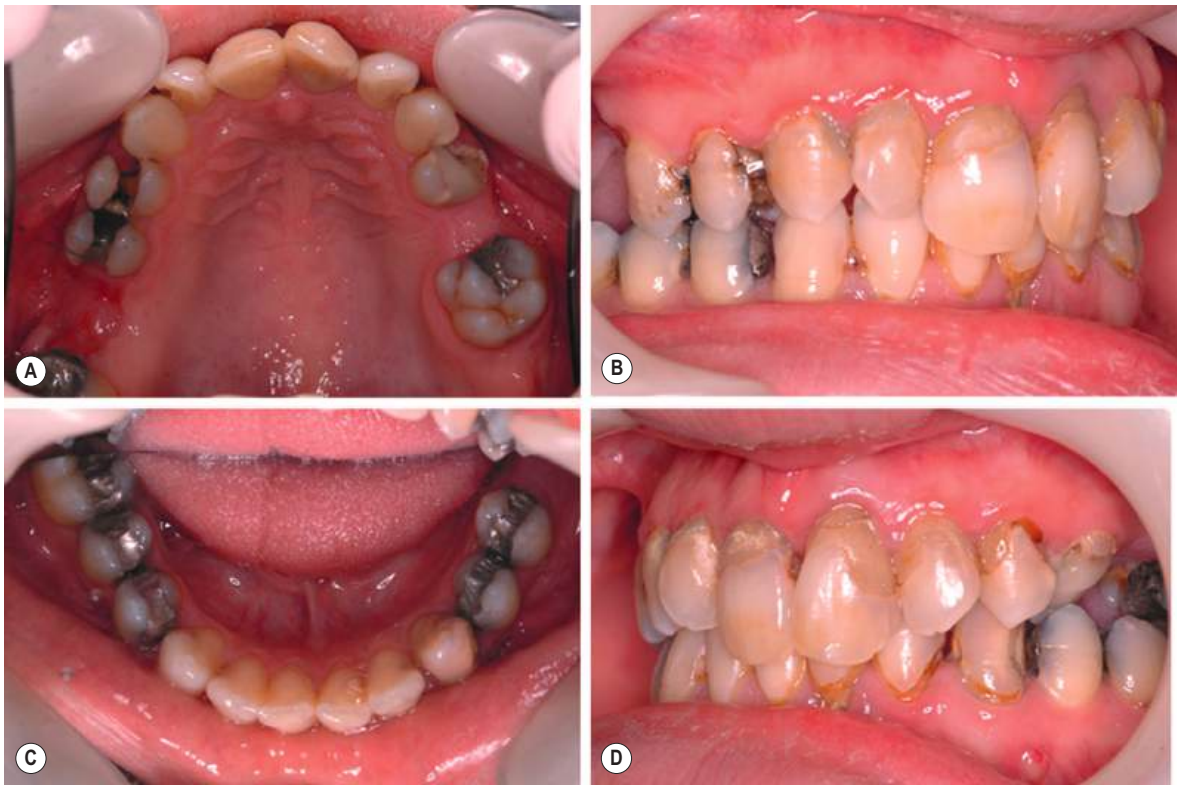


Figure 1.9 Clinical images of a high caries risk patient with multiple restorations and new carious lesions. The patient's bitewing radiographs are seen in Figure 1.4.

elimination of plaque-retentive factors such as poorly contoured restorations should take place. Reassessment of caries risk is then required over a period of time before advanced operative techniques are considered.

Caries risk factors

Social deprivation

Caries prevalence, and hence risk, has been shown to be closely associated with social deprivation in many countries. For example, in Scotland, DEPprivation CATegories (DEPCAT) have been derived from censuses in small post code areas. Variables taken into consideration in deriving DEPCAT scores are:

- Overcrowding
- Male unemployment
- Low social class
- No car.

DEPCAT score 1 is the most affluent and DEPCAT score 7 the least affluent. Figures 1.10 and 1.11 clearly demonstrate the relationship between caries prevalence and social deprivation, with the least affluent areas having fewer caries-free individuals and the highest average decayed (at the dentine level), missing and filled teeth ($d_3$mft). Whilst these figures are derived on a population basis, dentists need to be aware of this association, and careful assessment of caries risk on an individual patient basis needs to be carried out prior to advanced operative procedures.

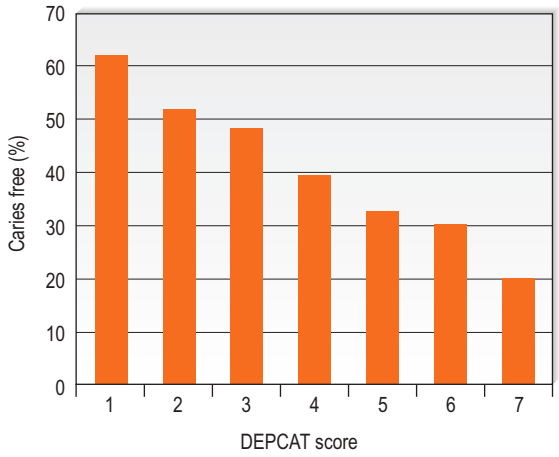


Figure 1.10 The relationship between social deprivation categories (DEPCAT score) and the percentage of the population who are caries free, the least affluent (score 7) having the fewest caries-free teeth for 5-year-old children.
(From Sweeney PC, Nugent Z, Pitts NB. Deprivation and dental caries status of 5-year-old children in Scotland. Community Dent Oral Epidemiol 1999;27:152–159.)

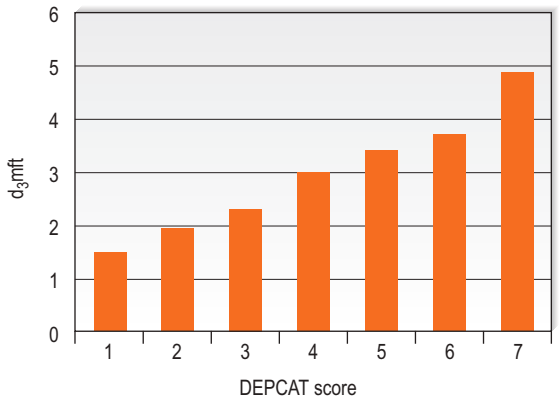


Figure 1.11 The relationship between social deprivation categories (DEPCAT score) and the percentage of the population who have dentine caries, missing or filled teeth ($d_3$mft), the least affluent (score 7) having the highest $d_3$mft for 5-year-old children.
(From Sweeney PC, Nugent Z, Pitts NB. Deprivation and dental caries status of 5-year-old children in Scotland. Community Dent Oral Epidemiol 1999;27:152–159.)

Past caries experience

A strong predictor of future caries is previous caries experience. A dental history and chart are important to assess dental attendance, how often restorations and re-restoration have been required, and whether teeth have been lost due to caries. The clinical and radiographic examination will reveal any new carious lesions. Following preventive advice, preventive treatment and stabilization of caries, assessment of the patient's compliance with attendance and treatment will be required. This will be followed by reassessment of caries risk over a period of 6–12 months before embarking on a more advanced phase of the treatment plan.

Oral hygiene

The level of oral hygiene can be assessed in a number of ways. At its simplest, the Silness and Löe index can be used to measure the amount and distribution of plaque accumulation around teeth. Commonly this is done on buccal and labial surfaces of undisclosed key teeth such as first molar, incisor and premolar teeth (e.g. UR6, UR1, UL4, LL6, LR1, LR6).

Silness and Löe plaque index:

- 0 = No plaque visible
- 1 = Plaque visible on a probe
- 2 = Plaque visible with the naked eye
- 3 = Plaque visible all around tooth.

This is useful as a screening tool, but to assess the level of oral hygiene throughout the mouth of a patient and to establish whether there has been a change in oral hygiene

with time, full mouth plaque scores should be recorded. For this, the tooth is divided into four surfaces: mesial, buccal, distal and lingual. The teeth are disclosed and the presence of plaque is recorded on a chart. The proportion of surfaces covered in plaque can then be calculated to give an objective figure at baseline with which comparison is made when reassessing caries risk. Figure 1.12 shows a typical plaque chart.

Many patients with poor oral hygiene may not have received any guidance on oral hygiene instruction and its implementation. Such guidance and implementation is a core feature in an effective preventive program. Disclosing plaque is an important first step for patients to see where the plaque builds up, where they are missing with their current oral hygiene procedures and to note the relationship between plaque accumulation and dental caries. The choice of toothbrush and appropriate tooth brushing technique must be discussed and demonstrated to the patient. Use of interdental aids, such as dental floss and interdental brushes, needs to be gradually introduced, so as not to overwhelm the patient.

It also is important when providing patients with fixed and removable prostheses that they are designed in such a way as to facilitate these oral hygiene procedures. Special additional hygiene procedures may need to be advised and then demonstrated; for example, the use of super floss, also known as three-in-one floss, beneath bridge pontics and connectors.

Diet

An association between dietary sugars and caries has long been established. Once sugar is consumed, the bacteria within the biofilm are able to produce acid, resulting in a rapid fall in plaque pH. When this falls below a critical pH, often considered to be in the region of pH 5.5, the plaque fluid becomes undersaturated with respect to tooth mineral, and demineralization of the tooth occurs. It may take some time for the pH and plaque fluid mineral saturation to return to resting levels. In terms of plaque and salivary pH this is characterized by the Stephan curve. A subsequent sugar snack may cause another dip in pH. Frequent sugar intakes may keep the biofilm undersaturated with respect to tooth mineral and below the critical pH for several hours each day. Sticky, sugary foods may also remain around the teeth for prolonged periods of time and have a similar effect.

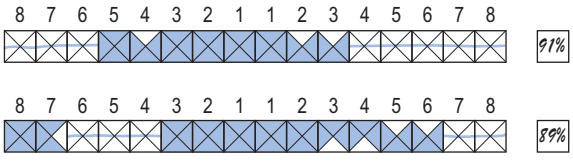


Figure 1.12 Typical plaque chart of a high caries risk patient before oral hygiene instruction and stabilization of caries.

In a caries risk assessment the diet needs to be evaluated. This can either be by patient recollection and appropriate questioning or a diet diary can be kept, in which all that is eaten and drunk for at least three consecutive days is recorded. This should include two working (or school) days and one non-work day because a patient's diet may be completely different on the different days. The diet diary can then be used to highlight the dietary sugar content, including any hidden sugars as well as the frequency of sugar consumption. This enables the practitioner to provide realistic and achievable diet advice. Figure 1.13 shows the diet diary of the patient seen in Figures 1.4 and 1.9. It is clear that there is frequent intake of sugar in coffee and tea, and mint sweets between meals. Either eliminating the sugar completely or using an artificial sugar substitute that is not cariogenic can be advised. At reassessment the diet diary can be repeated to assess compliance with this advice.

Diet diaries are frequently criticized as patients often do not fill them in correctly and can deliberately avoid recording what they know is likely to be the cause of the caries. It can be argued that if they know what not to include, they have the knowledge necessary to prevent their caries and may already be in the process of amending their diet.

Fluoride

Topical fluoride is important in a caries preventive program and information on its use needs to be gathered in a caries risk assessment. Although most European-marketed adult toothpastes contain fluoride at about 1400 ppm, some toothpastes do not contain fluoride, such as some obtained from health food shops and websites; if in doubt, check the list of ingredients. Frequent brushing with a fluoride toothpaste is adequate for lower caries risk individuals, but for those who are higher risk supplemental fluoride should be considered. These can be applied professionally in the surgery/office or by the patient at home. In the surgery/office, a fluoride varnish can be applied to early lesions and susceptible restoration margins. Duraphat varnish (Colgate-Palmolive (UK) Ltd, Guildford, Surrey) contains a 50 mg/ml suspension, which equates to 2.26% (22 600 ppm) of sodium fluoride. This should be avoided in patients who are hypersensitive to colophony, one of its constituents, and for patients with severe asthma who have been admitted to hospital for the condition. Application of fluoride in high doses leads to the formation of calcium fluoride which is relatively soluble and acts as a fluoride reservoir, protecting against further carious attack by inhibition of tooth mineral dissolution when the local pH falls. High dose fluoride toothpastes (Duraphat 2800 ppm F (not recommended for <10 year olds) and Duraphat 5000 ppm F (not recommended for <16 year olds); Colgate-Palmolive (UK) Ltd) are available by prescription and are useful for high-risk patients. Fluoride mouthwashes can be used on a daily (usually around 0.05% sodium fluoride, 227 ppm F) or weekly basis (usually around 0.2% sodium

	Sunday		Monday		Tuesday	
	Time	Items	Time	Items	Time	Items
Before breakfast	9.30	Mug of tea/1 sugar	6.45	Mug of tea/1 sugar		
Breakfast	10.30	Bacon roll Mug of tea/1 sugar	7.15	Bowl of porridge with syrup Mug of tea/1 sugar	7.15	Jam doughnut Mug tea/1 sugar
Morning	11.30 12.15	Mug of coffee/2 sugars, chocolate bar Half a packet of mints through morning	10.30	Mug of coffee/2 sugars, biscuit 3 mints in morning	10.45 12.00	Mug of coffee/2 sugars Mug of coffee/2 sugars
Mid-day meal	2.15	Bowl of soup Can of cola drink	1.00	Egg sandwich, bag crisps, chocolate bar Mug of coffee/2 sugars	1.30	Banana, crisps, chocolate bar Mug of coffee/2 sugars
Afternoon	4.00	Mug of coffee/2 sugars Crisps Half packet mints	3.30	Mug coffee/2 sugars Mints	 3.30 6.30	Mints Can of cola drink Mug tea/1 sugar
Evening meal	6.00	Roast chicken, veg, 2 glasses wine	8.30	Spaghetti bolognaise Can of cola drink	8.00	Prawns, salad, potatoes, water
Evening	8.00 10.00	Mug coffee/2 sugars Hot chocolate			9.30	Mug coffee/2 sugars

Figure 1.13 Diet diary of patient seen in Figures 1.4 and 1.9 at initial assessment.

fluoride, 909 ppm F) depending upon patient choice and what suits their lifestyle best. Note that formulations and fluoride content of mouthwashes may differ between countries.

Saliva

Saliva is extremely important for oral health and function. In its absence, there can be devastating effects, which include difficulty with mastication, swallowing and speech, loss of taste, oral soreness, a feeling of thirst and widespread rampant caries. The feeling of a dry mouth, or xerostomia, is usually a result of hyposalivation or oral dryness. It is a side effect of commonly prescribed drugs such as antihypertensives, diuretics, antidepressants, antipsychotics, antispasmodics and some antihistamines. Other causes of dry mouth include autoimmune diseases such as Sjögren's syndrome, diabetes, radiotherapy to the head and neck, and the use of recreational drugs such as caffeine, alcohol and amfetamines.

Clinically, a dry mouth can be obvious and an examination difficult as the dental mirror sticks to the oral mucosa. The amount of saliva production can also be measured under either unstimulated or stimulated conditions. The normal unstimulated and paraffin-wax-stimulated flow rates are given in Table 1.2, together with those for patients diagnosed as having hyposalivation.

Saliva and its constituents have many important properties which are illustrated in Figure 1.14. When providing

Table 1.2 Unstimulated and paraffin-wax-stimulated whole saliva for a normal patient and one with hyposalivation

SALIVA FLOW	UNSTIMULATED ml/min	STIMULATED ml/min
Normal	0.3	1.5
Dry mouth	≤0.1	≤0.5♀ : 0.7♂

patients with advanced operative dentistry it is important to bear in mind that these restorations have to be maintained throughout life and changes in a patient's medical history (e.g. medication causing a dry mouth) may increase their caries risk and hence the level of maintenance required in the future.

A patient with a dry mouth will want to take frequent sips of liquid. It is obviously important for this liquid not to contain sugar due to the reduced oral clearance. Plain water should be advised. Some patients also suck on sweets, especially those with a bitter taste, to stimulate some salivary flow. Again this should be avoided and sugar-free sweets or sugar-free chewing gum can be suggested as a good alternative. Even then care must be taken as some artificial sugars such as sorbitol can be metabolized by plaque in some subjects and this could be of

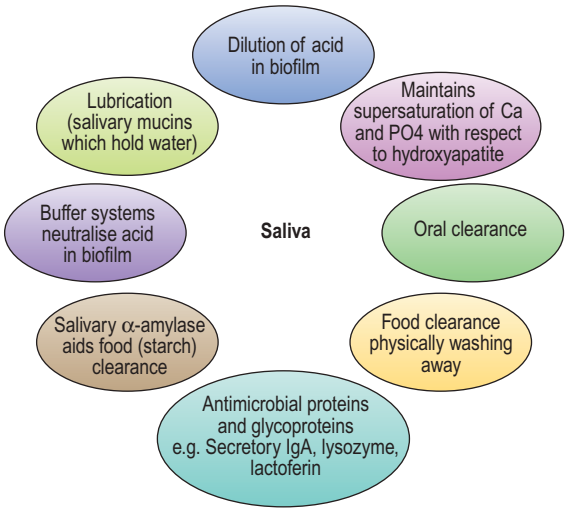


Figure 1.14 Important constituents and properties of saliva.



Figure 1.15 Commercially available kit to measure the counts of organisms within saliva (CRT Bacteria, Ivoclar Vivadent, Schaan, Liechtenstein).

concern for some xerostomic patients. For some patients, artificial saliva may give relief from the symptoms of dry mouth, but it is important to ensure that the pH is not acidic which could cause erosion in the dentate patient. Referral to an oral medicine specialist should also be considered for a patient with a dry mouth as a result of radiotherapy or Sjögren's syndrome. If there is sufficient functional glandular tissue remaining, prescription of pilocarpine can be considered to stimulate salivary flow.

Saliva microbiology

Salivary tests for mutans streptococci and lactobacilli have been used for many years and commercially available kits have been produced to measure the counts of both organisms within saliva. It is assumed that if the levels of these cariogenic organisms are high in the dental biofilm on the surface of the tooth or within active carious lesions, the levels will also be high in the saliva. Indeed, a number of studies have found associations with these two organisms in saliva and the caries experience of individuals. More recent evidence suggests that these salivary counts are not good in predicting future caries; however, they may be useful in assessing patients' compliance with dietary advice, for as the level and frequency of sugar in the diet reduces, the ensuing modification in the local oral environment is reflected in a reduced count of both species. This can serve as a tangible reward to a compliant patient and spur on their efforts in changing their dietary lifestyle.

Figure 1.15 illustrates one such kit (CRT Bacteria, Ivoclar Vivadent, Schaan, Liechtenstein). In this a sample of stimulated mixed saliva is collected and applied, at the chairside, to selective culture plates. Following incubation at 37°C for 48 hours, the numbers of colony forming units (CFUs) are estimated by comparing the cultures to a chart (Figure 1.16). Figures 1.17 and 1.18 illustrate this for a high caries risk patient and one of low risk.

The various caries risk factors described are subjectively drawn together for each patient, with the outcome influencing the patient's treatment plan. A more formal and objective way of assessing this is by using a computer-based caries risk model. The Cariogram is such a system which was developed in Malmö Dental School by Douglass Bratthall and co-workers. Information is gathered from the patient about caries risk, clinical and radiographic findings, and the results from various salivary tests. This information is given a score of 0–3 (or in some cases 0–2). These scores are then entered into the Cariogram program, from where information is weighted according to its impact on caries risk. The program evaluates the data and then presents it as a pie chart with a clear indication of future caries risk expressed as a 'Chance to avoid caries'. The various risk factors described above and those evaluated in the Cariogram and the corresponding scores given are detailed in Table 1.3. Table 1.3 also summarizes actions that can be taken to address each factor positively, with an aim to change the patient's caries risk. Figures 1.19 and 1.20 show what the Cariogram would look like for a low- and a high-risk patient, respectively.

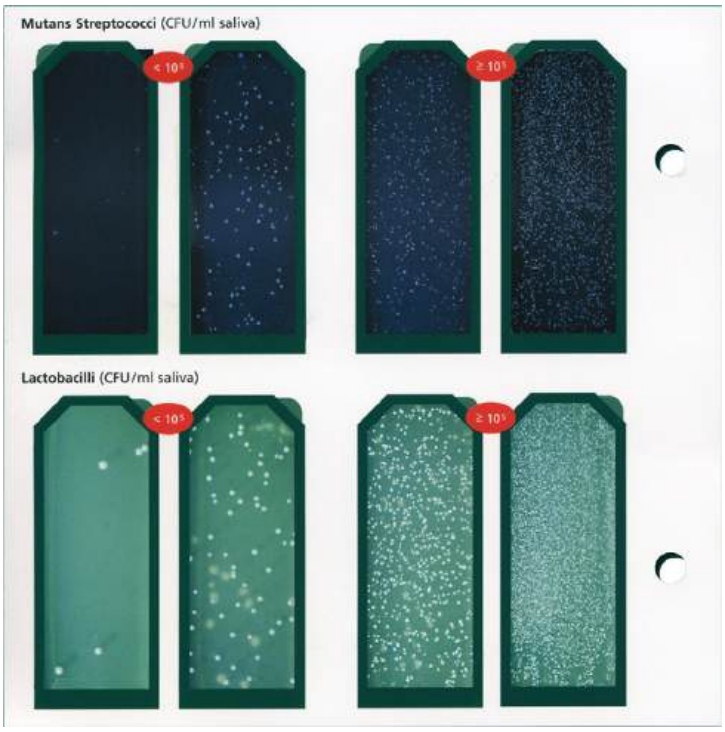


Figure 1.16 Chart enabling the number of colony forming units (CFUs) for mutans streptococci and lactobacilli to be estimated.

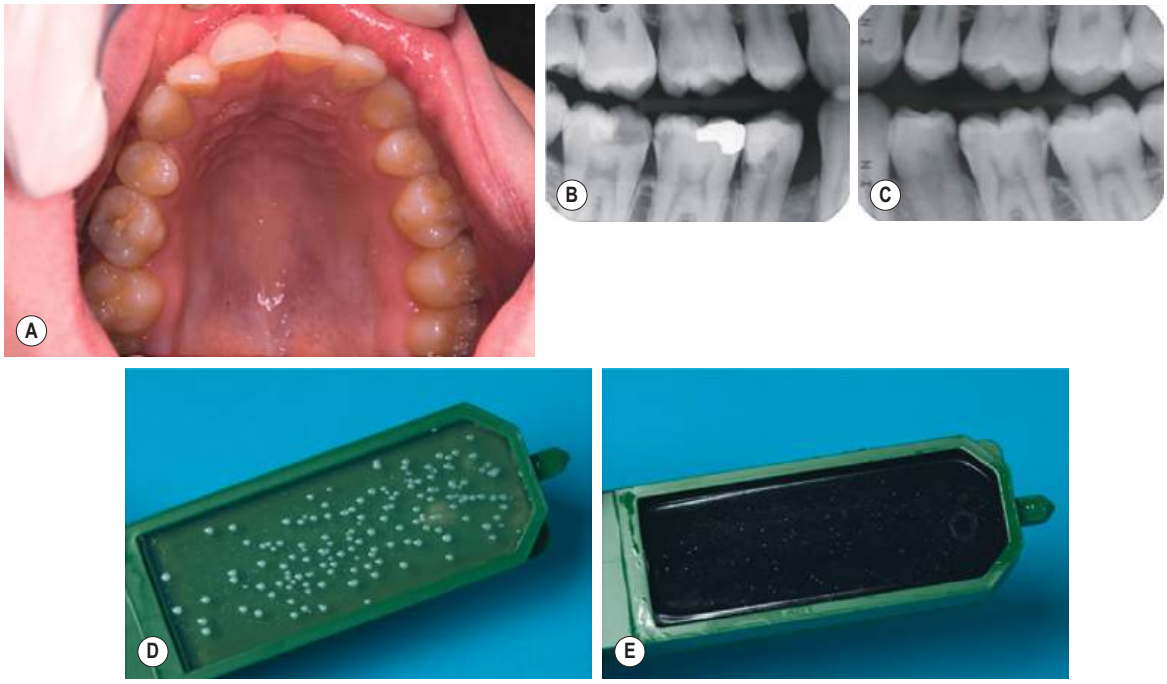


Figure 1.17 Clinical image of upper arch and bitewing radiographs of a high caries risk patient together with their lactobacilli and mutans streptococci counts.

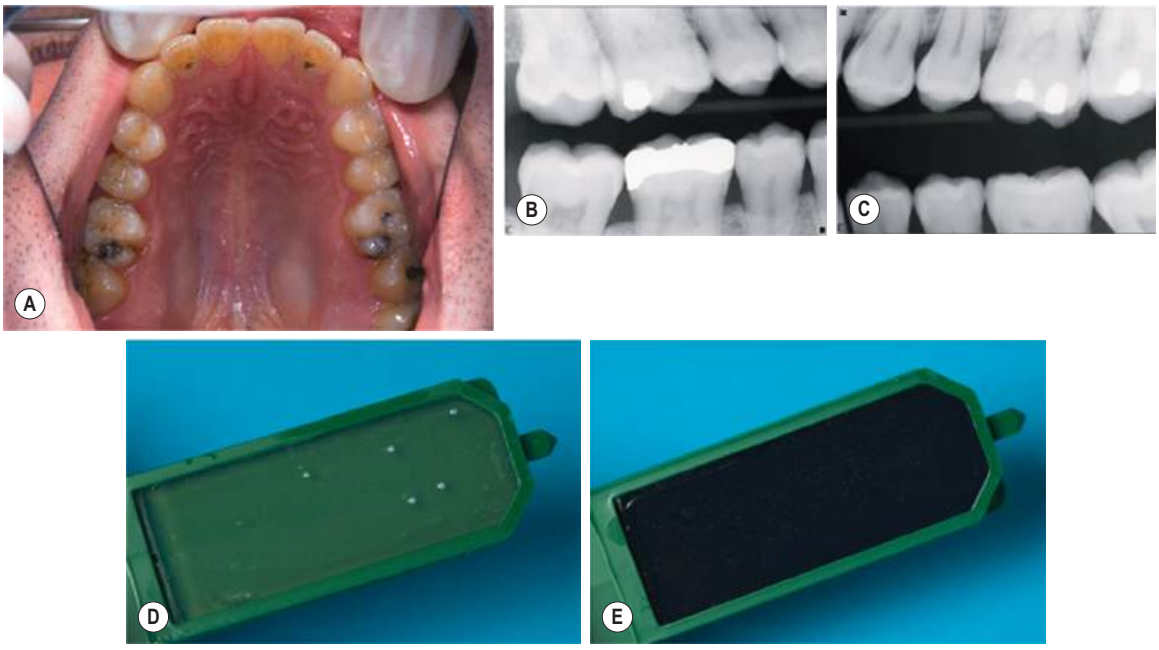


Figure 1.18 Clinical image of upper arch and bitewing radiographs of a low caries risk patient together with their lactobacilli and mutans streptococci counts.

Table 1.3 Examples of various caries risk factors, how they can be assessed, the score they would be given in a caries risk model (the Cariogram) and how that risk factor can be modified

FACTOR	INFORMATION TO BE COLLECTED	CARIOGRAM SCORE	ADVICE AND MANAGEMENT
Caries experience	Decayed/Missing/Filled Teeth (DMFT), Decayed/Missing/Filled Surfaces (DMFS), new caries in last year	0: No caries, no fillings 1: Better than normal for age group 2: Normal for age group 3: Worse than normal for age group	All of the below
Related general diseases	Medical history and medications	0: Healthy 1: Illness that contributes moderately to caries risk 2: Serious illness with strong influence on caries	Liaise with medical practitioner – avoid medication with sugar or causes dry mouth where possible
Diet content	Diet history (or *Lactobacillus* test), quality of diet	0: Very low sugar content (LC $\leq$1000) 1: Low sugar content (LC 10^4) 2: Moderate sugar content (LC 10^5) 3: High sugar content (LC $\geq 10^6$)	Reduce sugar consumption Suggest safe snacks and drinks Artificial sweeteners
Diet frequency	Diet history – mean number of meals/ snacks per day	0: Maximum 3 meals/day 1: Maximum 5 meals/day 2: Maximum 7 meals/day 3: More than 7 meals/day	Advice on reduced frequency of sugar intake Sugar with main meal Safe snacks

(Continued)

Table 1.3 Examples of various caries risk factors, how they can be assessed, the score they would be given in a caries risk model (the Cariogram) and how that risk factor can be modified—cont'd

FACTOR	INFORMATION TO BE COLLECTED	CARIOGRAM SCORE	ADVICE AND MANAGEMENT
Amount of plaque	Silness–Löe plaque index	0: No plaque visible 1: Plaque visible on a probe 2: Plaque visible with the naked eye 3: Plaque visible all around tooth	Disclose, oral hygiene instruction, toothbrush technique, electric toothbrush with oscillating head, interdental aids (interdental brushes and floss)
Streptococcus mutans	Salivary mutans streptococci counts	0: *S. mutans* <104/ml saliva 1: *S. mutans* <106/ml saliva 2: *S. mutans* <107/ml saliva 3: *S. mutans* >107/ml saliva	Improved oral hygiene (see above)
Fluoride programme	Estimation of extent of fluoride exposure	0: Maximum fluoride exposure 1: Additional fluoride measure (other than toothpaste) but infrequent application 2: Fluoride toothpaste only 3: Avoidance of fluorides (e.g. no fluoride exposure)	Ensure brush at least 2–3 times per day with at least 1400 ppm fluoride toothpaste High-risk supplement with professional application of fluoride varnish, high-dose fluoride toothpaste or fluoride mouthwash
Saliva secretion rate	Stimulated saliva flow rate	0: ≥0.7 ml saliva/min 1: 0.3–0.7 ml saliva/min 2: ≤0.3 ml saliva/min	Liaise with medical practitioner – consider alternative medication (less of a dry mouth side effect) if possible Dietary advice – sugar-free drinks, chewing gum, sweets Artificial saliva Oral Medicine referral? Prescription of pilocarpine
Saliva buffering capacity	Buffer kit – Dentobuff	0: Adequate, saliva pH >6.0 1: Reduced, saliva pH 4.5–5.5 2: Low, saliva pH <4.0	

Table generated from Alian et al. (2006), Bratthall et al. (2001) and Ruiz Miravet et al. (2007) LC, Lactobacillus Counts.

SUMMARY AND PRINCIPLES OF TREATMENT PLANNING FOR A HIGH CARIES RISK PATIENT

Early caries detection and caries risk assessment are important to target prevention for those patients that are at risk of developing new caries. Monitoring of caries is essential to determine the success of preventive care. When this primary prevention has not been instituted or has failed, a clinical outcome such as that described in Figures 1.4 and 1.9 may be observed. History taking, examination, diagnosis and treatment planning for this patient should be structured and staged in the following way and allow re-evaluation prior to complex and advanced restorative dentistry:

- Any emergency treatment should be provided first; this may include extraction or extirpation of a pulp for a tooth with an irreversible pulpitis.
- Assessment of risk factors.
- Appropriate preventive advice and treatment should be given as outlined in Table 1.3 based upon the individual's risk factor.
- Stabilization of disease. In relation to gross caries, a quadrant at a time may be taken and caries removed from all teeth in that quadrant and provisionally restored. For deep carious lesions, stepwise excavation can be considered (see Ricketts, 2001). If root canal treatment is needed for any teeth, the first stage of treatment (e.g. pulpal extirpation) can be carried out and the tooth provisionally restored.
- Each tooth can then be carefully and definitively restored, ensuring good adaptation, cavity seal and

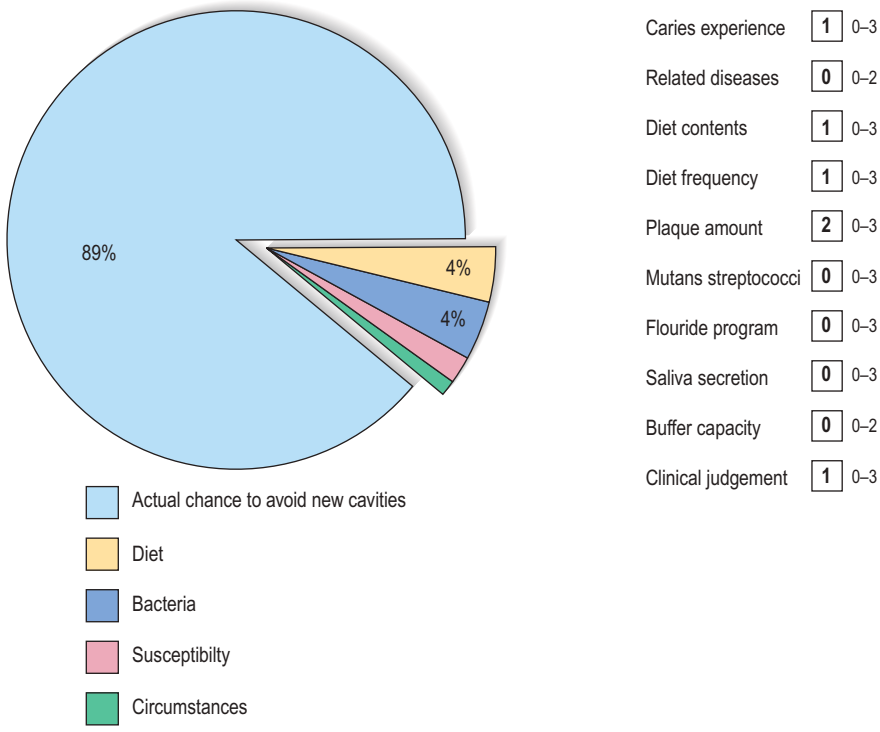


Figure 1.19 Cariogram of a low caries risk individual with a high (89%) chance of avoiding new caries.

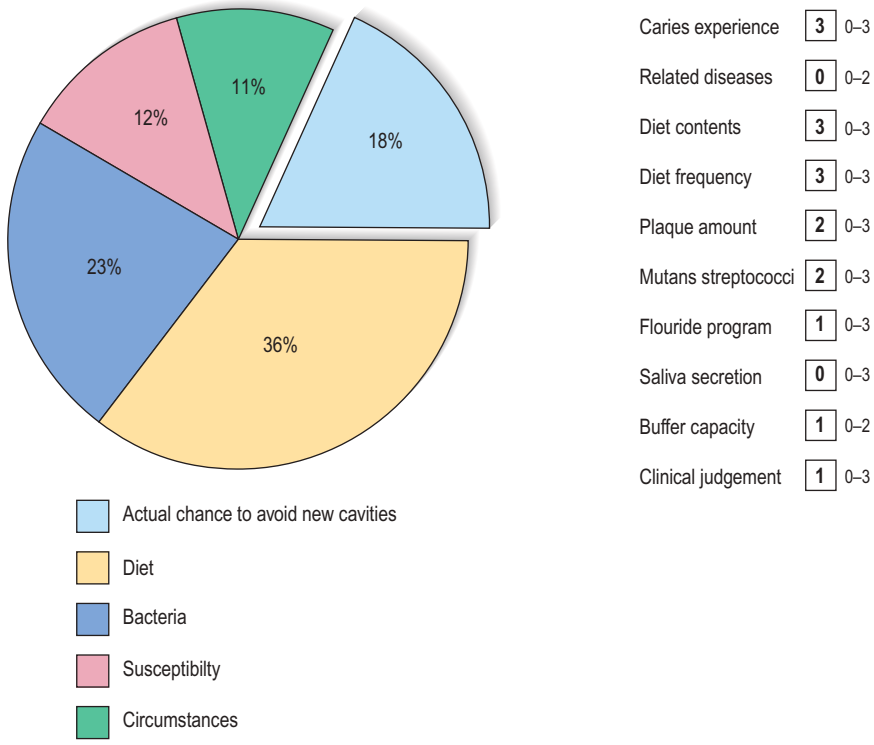


Figure 1.20 Cariogram of a high caries risk individual with a low (18%) chance of avoiding new caries.

contour of the restorative material chosen. Initial stabilization will ensure that other lesions do not progress whilst this protracted stage of the treatment plan is being carried out – continued preventive advice and support should be conducted throughout.

- Once all the provisional restorations have been replaced, consideration can be given to endodontic treatment of those teeth that require it.
- Reassessment of caries risk. Re-evaluation of all the risk factors in Table 1.3 should now take place and any new lesions recorded. If the patient appears to have addressed the main risk factors and no new lesions have appeared, consideration can now be given to an advanced stage of the treatment plan, which might include crowns and bridges. If in any doubt as to the patient's caries risk status and compliance with preventive advice, a further period of monitoring would be advised.

FURTHER READING

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Bratthall, D., Stjernswärd, J.R., Petersson, G.H., 2001. Assessment of caries risk in the clinic – a modern approach. In: Wilson, N.H.F., Roulet, J.F., Fuzzi, M. (Eds.), Advances in operative dentistry: challenges of the future. Quintessence Publishing, London, pp. 61–72.

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